Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions

Rows in dataset

18

Columns

Features tracked

37

Missing Values

Review Rating only

$59.76

Avg. Purchase

Mean spend per order

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Key Features Tracked

Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Promo Code, Frequency, Review Rating, Shipping

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Data Preparation in Python

01

Load & Explore

Imported via pandas; `df.info()` & `.describe()`

02

Clean Missing Data

Imputed Review Rating using category medians

03

Feature Engineering

Created `age_group` & `purchase_frequency_days`

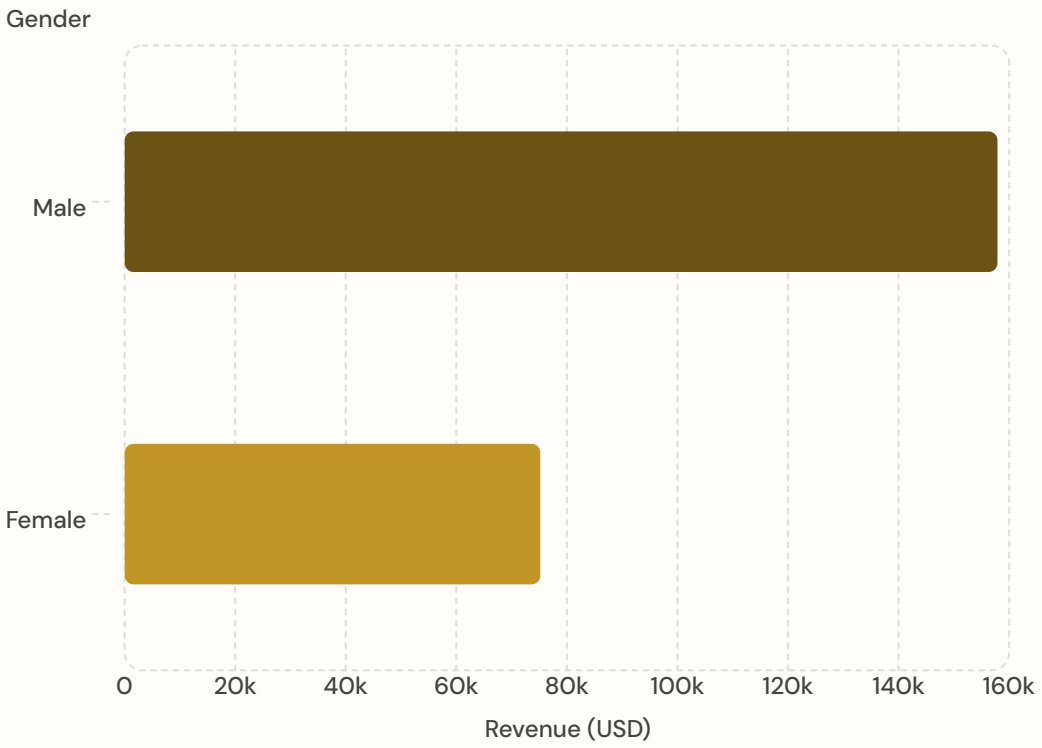
04

Database Integration

Loaded cleaned DataFrame into PostgreSQL

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Revenue by Gender



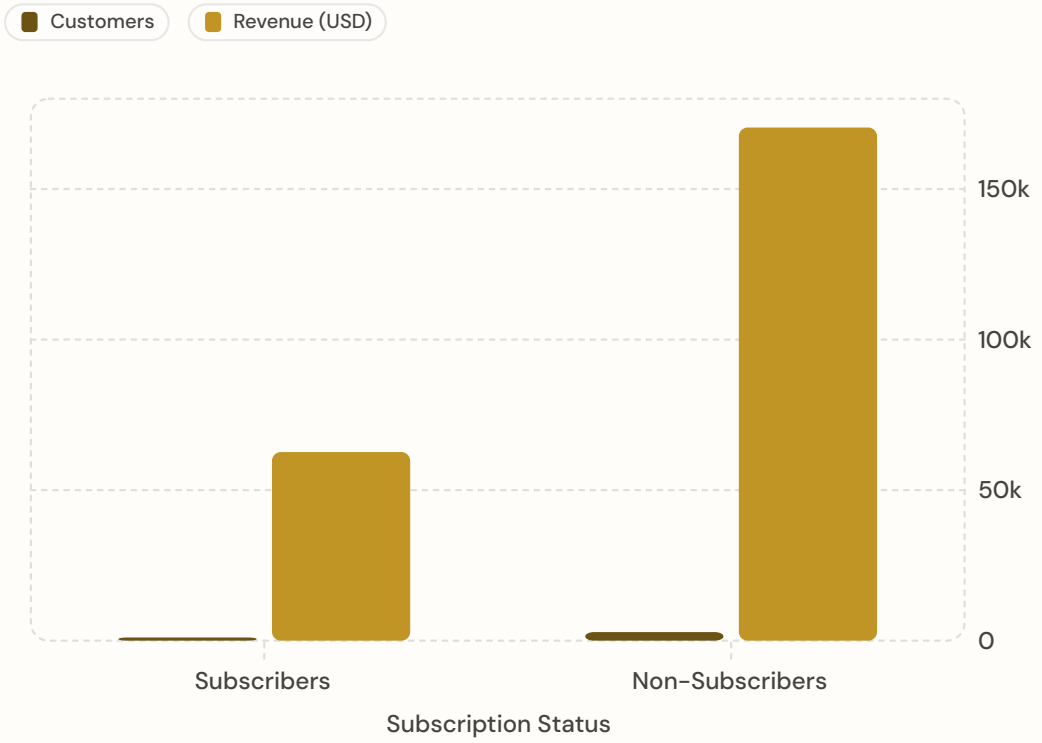
Key Insight

Male customers generated **2.1× more revenue** than female customers.

- Male: **$157,890**
- Female: **$75,191**

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Subscriber vs. Non-Subscriber



Key Insight

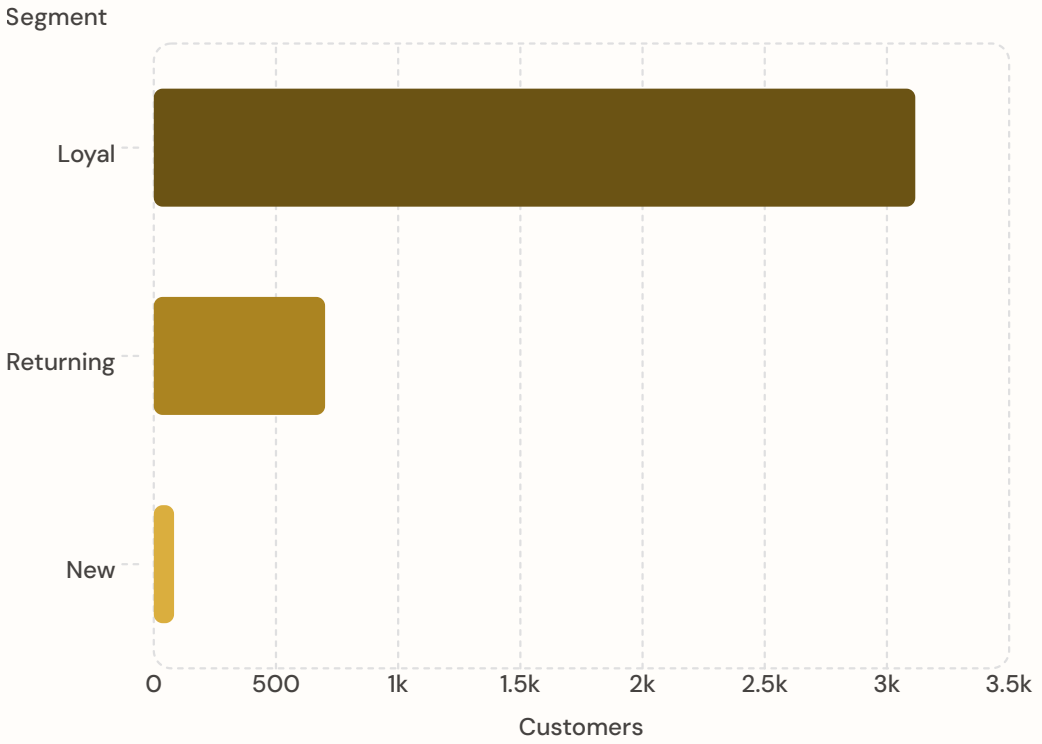
Non-subscribers drive **73% of total revenue** despite similar avg. spend.

- Avg spend: ~$59.50 (both groups)
- Only **27%** are subscribers

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Customer Segmentation & Top Products

Customer Segments

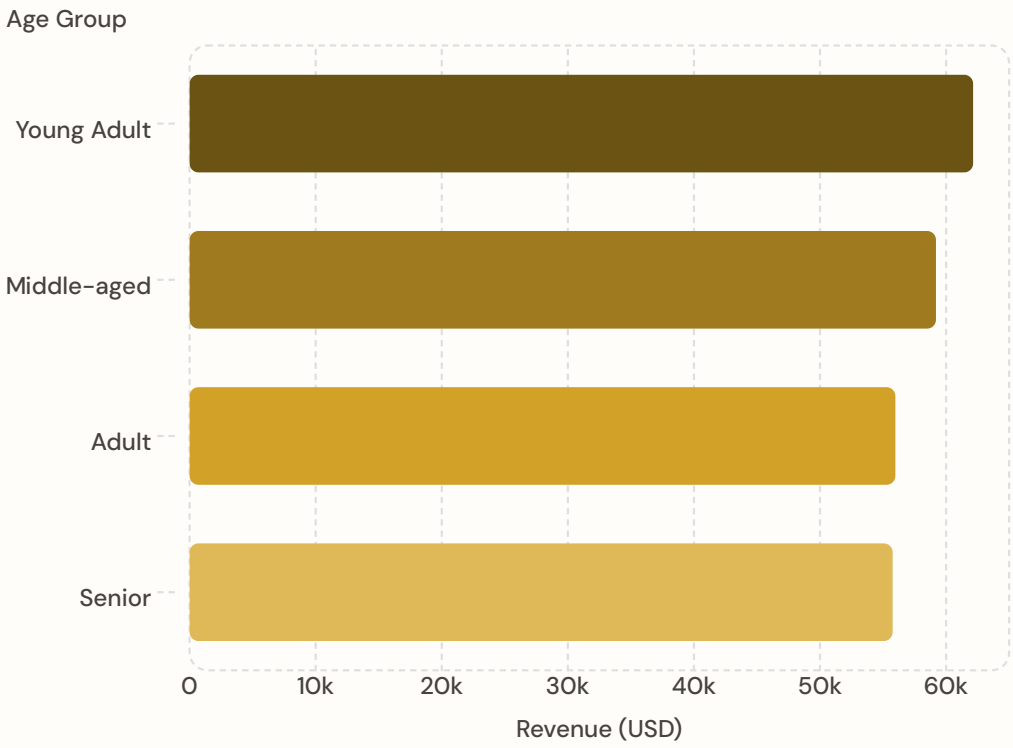


Top-Rated Products

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

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Revenue by Age Group



Key Insight

Revenue is **relatively balanced** across age groups — Young Adults lead slightly.

- Young Adult: **$62,143**
- Senior: **$55,763**

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Power BI Dashboard

3.9K Customers

Total tracked

$59.76 Avg.

Purchase amount

3.75 Avg.

Review rating

73% Non-Subscribers

Largest segment

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Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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